

Ethan Lo

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EDUCATION

Georgia Institute of Technology, Atlanta, Georgia

May 2028

Bachelor of Science in Computer Science, Threads: AI and Modeling & Simulation

GPA: 3.77/4.00

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Computer Organization & Programming, Startup Lab, Calculus I & II, Linear Algebra, Multivariable Calculus, Applied Combinatorics, Differential Equations

TECHNICAL SKILLS

ML/AI – XGBoost, GradientBoost, Logistic Regression, KNN, RF, MLP, SVM, CNN, RNN, LSTM, BERT

Languages & Frameworks – Python, Java, JUnit, HTML/CSS, JavaScript

Databases & Tools – VS Code, IntelliJ, Git, Jupyter Notebook, Google Colab, Raspberry Pi

WORK EXPERIENCE

ScoutOut via Harvard Ventures Technology Summer Program, Remote – AI/ML Intern

Jun 2025 – Jul 2025

- Developed a regression model trained on 50,000 properties to predict real estate value from location and market features
- Predicted with a MAE of 0.0273 and R^2 score of 0.941 using XGBoost
- Presented interactive UI demo and analytical results to company founder (ex-Amazon)

RESEARCH PROJECTS

Generation of Folksong Melodies with Deep Learning and Valence-Arousal Emotions

Aug 2025 – April 2026

- Proposed a generative LSTM model trained on the Essen Folksong Database to create MIDI sequences of melodies
- Collected participants' emotional perception of generated melodies via online anonymous surveys
- Achieved a validation loss of 3.99; melodies demonstrated pitch/rhythmic diversity and convincing human character

Cyclone Trajectory Prediction and Status Classification with ML/DL Techniques

Sep 2025 – Nov 2025

- Applied GradientBoost, RF, SVM, and MLP techniques to forecast cyclone trajectories and classify storm intensity statuses
- Optimized algorithms for time-series weather data using sliding windows; best accuracy achieved: 93%
- Managed data preprocessing, model build, and manuscript composition (arXiv: <https://doi.org/10.48550/arXiv.2509.24146>)

Exoplanet Detection ML Model Trained on Synthetic Light Curves

Jul 2024 – Mar 2025

- Conducted research on NASA Kepler telescope data of 5,000+ exoplanet candidates and synthetic light curves using SMOTE
- Directed a 5-person team for data preprocessing and model design using logistic regression, KNN, and RF
- Achieved 92% classification accuracy; independent manuscript published (arXiv: <https://doi.org/10.48550/arXiv.2507.19520>)

PROGRAMMING PROJECTS

Mental Health Sentiment Analysis Platform

April 2025

- Collaborated with a 4-member team to build an NLP-based mental health sentiment analysis platform using BERT
- Created a React website to demonstrate functionality with data visualization tools for emotional awareness
- Earned 1st Place at the Empower Atlanta Divergent Teams Hackathon at Microsoft among 13 Georgia teams

Object Detection Using a Resource-Limited Device with Machine Learning

Jan 2024

- Built an object detection system backed by YOLOv5 supporting both image and video stream inputs
- Deployed on a Raspberry Pi with low computational overhead, achieving 20 FPS
- Received 3rd Place at the North Cobb Technology Competition in the programming category

LEADERSHIP AND COMMUNITY ENGAGEMENT

Vice President, Walton High School Computer Science Honor Society

Aug 2025 – May 2026

- Established coding tutoring programs at local elementary schools
- Promoted CS education and organized hackathons and seminars at Walton

Founder, Walton Public Speaking Club

Aug 2025 – May 2026

- Empowered 20+ students in oral excellence and effective communication at biweekly meetings
- Organized the first TEDxWaltonHigh event with 5 speakers for 50+ attendees

Volunteer, Overseas Community Affairs Council of Taiwan English Teaching Volunteer Service Program

July 2026

- Completed teacher training at Taiwan Medical University
- Taught English to elementary students at rural locations across Taiwan for 3 weeks

ADDITIONAL SKILLS AND INTERESTS

Languages: English, Mandarin, Latin, Spanish

Interests: Tennis, volleyball, cello, calisthenics